

MINI ASSESSMENT

Rule-Based Python Chatbot

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Programming Language: Python

1. Objective

To develop a simple rule-based chatbot in Python that accepts user messages and provides predefined responses based on the entered keywords.

2. Concept

A rule-based chatbot works using predefined rules. The program checks the user's input and selects a matching response. It does not learn from conversations; its responses are written in advance by the programmer.

Basic flow: **User Input** → **Rule Matching** → **Predefined Response**

3. Working

The chatbot continuously takes input from the user. It recognizes common messages such as greetings, questions about its name, requests for help, and thanks. When the user enters bye or exit, the chatbot stops.

Python Implementation

The following program implements the rule-based chatbot using simple if/elif conditions.

```
def chatbot():
    print("Rule-Based Python Chatbot")
    print("Type 'bye' or 'exit' to stop.")

    while True:
        user = input("You: ").lower().strip()

        if user in ["hello", "hi", "hey"]:
            print("Bot: Hello! How can I help you?")

        elif "name" in user:
            print("Bot: I am a rule-based Python chatbot.")

        elif "help" in user:
            print("Bot: I can answer simple predefined questions.")

        elif "thank" in user:
            print("Bot: You're welcome!")

        elif user in ["bye", "exit"]:
            print("Bot: Goodbye!")
            break

        else:
            print("Bot: Sorry, I don't understand that.")

chatbot()
```

4. Key Features

- Accepts continuous user input
- Recognizes predefined messages
- Gives predefined responses
- Handles unknown input
- Exits using bye or exit

Execution and Conclusion

5. Sample Execution

Rule-Based Python Chatbot

Type 'bye' or 'exit' to stop.

You: hi

Bot: Hello! How can I help you?

You: what is your name?

Bot: I am a rule-based Python chatbot.

You: help

Bot: I can answer simple predefined questions.

You: thank you

Bot: You're welcome!

You: xyz

Bot: Sorry, I don't understand that.

You: bye

Bot: Goodbye!

6. Conclusion

The Rule-Based Python Chatbot demonstrates how a basic conversational program can be created using Python conditions and predefined responses. User input is matched with specific rules, and the chatbot produces the corresponding response. This is a simple introduction to chatbot development.